Project Proposal

DEPI · Project Planning & Management · UK Railway Journeys Analysis: Revenue, Delays & Refunds Dashboard

31,653	4	6	6	5
Ticket transactions	Months of journeys	Business questions	Team members	Tools

Project information

Project title	UK Railway Journeys Analysis: Revenue, Delays & Refunds Dashboard	**Program**	DEPI, Digital Egypt Pioneers Initiative
Team	OMNI	**Track**	Data Analysis
Team leader	Nada Ahmed El-Sayed	**Instructor**	Ahmed Ayman
Team members	Saja Ahmed Saeed, Omar Saad Wahba, Habiba Islam Mohamed, Jenny Mohamed Hassan, Mostafa El-Sayed Abdelaziz	**Date**	16 October 2026

1. Overview

This project analyzes **31,653 UK train ticket transactions** (journeys from January to April 2024) and delivers an interactive **Power BI dashboard** covering ticket revenue, journey performance, delay causes, and refund requests.

2. Problem statement

About **13%** of journeys in the dataset were either delayed or cancelled, and **1,118** passengers requested a refund. The cause of a disruption is also recorded inconsistently, and there is no ready view showing where, when, and why journeys fail.

Without this view, management cannot decide which routes, stations, and causes to fix first, or understand how disruptions affect revenue and refunds.

7.2%	5.9%	1,118
Delayed journeys	Cancelled journeys	Refund requests

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3. Objectives

#	Objective
1	Clean and standardize the data and document every step.
2	Build a reproducible ETL pipeline using Python, Power Query (M Language), Databricks (SQL), and Power BI.
3	Design a Star Schema for the data.
4	Define KPIs: Total Revenue, On-Time %, Delay %, Cancellation %, Average Delay Minutes, and Refund Rate.
5	Deliver an interactive Power BI dashboard that answers the business questions.
6	Turn the findings into clear, data-backed recommendations.
7	(Optional stretch goal) If time allows and the approach works well, build a baseline machine learning model that estimates the risk of a journey being delayed or cancelled. The team has not committed to this yet.

4. Business questions

#	Business question	Theme
Q1	Which routes and stations have the most delays and cancellations?	Performance
Q2	What are the main causes of delays and cancellations, and how do they change over time?	Causes
Q3	How much revenue is linked to disrupted journeys and refund requests?	Revenue
Q4	How do ticket type, ticket class, and railcard affect revenue?	Revenue
Q5	When are the peak purchase and travel times?	Demand
Q6	Do online and station purchases behave differently?	Demand

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5. Data

Item	Detail
Source	UK railway ticket dataset (`railway.csv`) with a data dictionary
Size	31,653 rows x 18 columns (one row = one ticket)
Journey dates	1 January to 30 April 2024
Stations	12 departure stations, 32 destinations
Content	Purchase details, ticket and price, journey details, and journey performance (status, delay reason, refund request)

6. Scope

In scope

- Journeys from January to April 2024
- Data cleaning, transformation, and loading (ETL)
- Star Schema, KPIs, and DAX measures
- Interactive Power BI dashboard
- Insights and recommendations
- Documentation and source code on GitHub

Stretch goal

- Possible baseline ML model for disruption risk, built in Python
- Not committed: only if there is enough time and the approach works well for the team

Out of scope

- Real-time data
- Production deployment of the ML model
- Actual refund amounts (not available in the data)

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7. Proposed approach



Step	Tool	Purpose
Cleaning	Python (Pandas)	Standardize values, fix data types, validate the data
Transformation	Power Query (M Language)	Clean, structure, and prepare the data for analysis (data types, date and time fields)
Storage and analysis	Databricks (SQL)	Load tables, validate, explore, answer the business questions
Dashboard design	Figma	Wireframes, layout, and user interface
Model, measures, dashboard	Power BI (DAX)	Data model, KPIs, and interactive visuals
Version control	GitHub	Code and documentation

Optional machine learning model: the team may build a baseline ML model (Python) if there is enough time and it works well. This will be decided after the SQL analysis is completed, based on the remaining time and on how the core work is going. The project is complete without it.

8. Timeline

Milestone	Deadline
Planning, literature review, requirements	16 Oct 2026
System analysis and design	6 Nov 2026
Implementation	30 Nov 2026
Final presentation, testing, and reports	4 Dec 2026

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9. Expected deliverables

Deliverable	Completion criteria
Cleaned dataset and Python cleaning script	Data is standardized and every cleaning decision is documented.
ERD and Star Schema design	The data model and relationships are clearly defined.
SQL scripts for validation and analysis	Queries and results address the business questions.
Power BI dashboard (.pbix)	Dashboard presents the KPIs and visuals, with working filters.
Project documentation	Plan, requirements, design, and test cases.
GitHub repository with README	Code and documents are organized and reproducible.
User manual and technical documentation	Explains how to use and maintain the solution.
Final presentation	The team can demonstrate the dashboard and explain the key findings.
Project infographic	A one-page visual summary of the project and its main results.
Machine learning model (optional)	Only if the team decides to build it; not required for project completion.